ALGEBRA 1 · UNIT 1: FOUNDATIONS OF ALGEBRA

Game Score Designer

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students become game designers for "Galaxy Dash." They write and evaluate scoring expressions, apply order of operations and integer rules to score cards, use properties to simplify bonus rounds, and design and balance a scoring system for their own game.

Expressions get meaning when every term is a game event. Students write, evaluate, and simplify expressions, interpret coefficients and factors, and check their order of operations and integer arithmetic on score cards. The design task asks them to balance a scoring system, which requires comparing expressions.

Standards Addressed

- A-SSE.1a Interpret parts of an expression (terms, factors, coefficients)
- A-SSE.1b Interpret complicated expressions by viewing parts as single entities
- A-SSE.2 Use the structure of an expression to rewrite it
- N-RN.3 Sums and products of rational and irrational numbers

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch: show a game screen and ask how the score is calculated. Part 1: write and evaluate the score expression.
Day 2	Part 2: score cards (order of operations, integers), bonus rounds (distributive property and like terms).
Day 3	Part 3: design a scoring system and pitch the game.

Materials

- Student handout
- Calculators (optional)
- Poster paper or slide template
- Optional: a free online game to show scoring

How the Three Versions Differ

Level	What changes
CP2	Expressions from the rules card with starters; evaluating with whole numbers and integers; one distributive bonus round.
CP1	Adds verbal phrases, an order-of-operations error analysis, combining like terms across levels, interpreting factors, and classifying real-number scores.
Honors	Generalizes scoring with letters for point values, simplifies nested expressions, proves expressions equivalent with named properties, and reasons about rational and irrational scores.

Watch For (Common Misconceptions)

- Subtracting before multiplying (200 − 4s evaluated left to right as (200 − 4)s).
- Distributing to only the first term in a bonus round.
- Combining unlike terms such as 10c + 50g = 60cg.

Extension Ideas

- Have students play-test each other's scoring systems.
- Add a multiplier power-up and write it with parentheses.
- Connect to Unit 2: what number of coins gives a score of exactly 500?

Answer Key – CP2

Part 1: Write the Score

#	Task	Answer
1	Write an expression for the points from c coins.	10c
2	Write an expression for a player's score from coins, gems, and hits (no time bonus yet).	10c + 50g − 25h
3	Find the score for a player with 12 coins, 3 gems, and 2 hits.	10(12) + 50(3) − 25(2) = 120 + 150 − 50 = 220 points
4	The time bonus is 200 − 4s. Find the time bonus for a player who used 30 seconds. What happens to the bonus as s gets bigger?	200 − 4(30) = 200 − 120 = 80 points. As s gets bigger, the bonus gets smaller.
5	In 10c + 50g − 25h, name the coefficient of g and explain what it means in the game.	50; each gem is worth 50 points.

Part 2: Check the Score Cards

#	Task	Answer
6	A score card says 3(10 · 4 + 50) − 25 · 2. Evaluate it using the order of operations.	3(40 + 50) − 50 = 3(90) − 50 = 270 − 50 = 220
7	In one round a player's points changed by +50, −25, −25, +10, and −40. What is the total change?	50 − 25 − 25 + 10 − 40 = −30 points
8	Double Points round: the score is 2(10c + 50g − 25h). Use the distributive property to rewrite it without parentheses.	20c + 100g − 50h
9	Explain why 10c + 50g = 50g + 10c. Which property is this?	Commutative property of addition: you can add in any order.

Part 3: Design Your Game

#	Task	Answer
10	Name your game and list at least three scoring events, including one penalty. Choose a point value for each.	Answers vary.
11	Write your score expression. Find the scores for two different players.	Answers vary.
12	Add a bonus round that multiplies the score. Rewrite it with the distributive property.	Answers vary.

Answer Key – CP1

Part 1: Write the Score

#	Task	Answer
1	Write the full score expression, including the time bonus.	10c + 50g − 25h + 200 − 4s
2	Translate: "Level 2 coins are worth 5 more than Level 1 coins, gems are worth 10 more, and hits cost 5 more." Write the Level 2 score expression...	15c + 60g − 30h
3	Write and simplify the total score for a player who collects the same c, g, and h on Level 1 and Level 2.	(10c + 50g − 25h) + (15c + 60g − 30h) = 25c + 110g − 55h
4	Find the full Level 1 score for c = 15, g = 2, h = 4, s = 45.	150 + 100 − 100 + 200 − 180 = 170 points

Part 2: Check the Score Cards

#	Task	Answer
5	Kai evaluated the time bonus $200 - 4(15 - 3)^2 \div 8$ and got −88. Find the correct value and explain Kai's mistake.	$(15 - 3)^2 = 144$; $4(144) = 576$; $576 \div 8 = 72$; $200 - 72 = 128$. Kai multiplied before squaring: $(4 \cdot 12)^2 = 2{,}304$; $2{,}304 \div 8 = 288$; $200 - 288 = -88$. Exponents come before multiplication.
6	A team score is 4(10c + 50g) − 25h. What could the 4 represent? What does (10c + 50g) represent?	4 could be the number of players on a team (each earning the same coins and gems). (10c + 50g) is the points one player earns from coins and gems.
7	Simplify 3(2c − g) + 2(c + 4g) − 5g and interpret the result as a scoring rule.	6c − 3g + 2c + 8g − 5g = 8c. Only coins count; gems cancel out!
8	The studio adds a "pi power-up" worth π points and a half-gem worth 2.5 points. Classify each possible score as natural, whole, integer, rational,...	220: natural, whole, integer, rational. −30: integer, rational. 2.5: rational. 0: whole, integer, rational. 230 + π: irrational.

Part 3: Design & Balance Your Game

#	Task	Answer
9	Write a score expression with at least three terms (one penalty) and a time bonus.	Answers vary.
10	Describe two different players (strategies) and show with evaluation that they earn the same score.	Answers vary.

11	Add a Level 2 or bonus round. Write the total score and simplify it. Explain one term of the simplified expression.	Answers vary.

Scoring note: For the Kai item, look for the idea that the exponent applies only to $(15 - 3)$, not to $4(15 - 3)$.

Answer Key – Honors

Part 1: General Scoring

#	Task	Answer
1	Write a general score expression S for c coins, g gems, and h hits. Then write Galaxy Dash's score by substituting p = 10, q = 50, r = 25.	S = pc + qg − rh; S = 10c + 50g − 25h
2	Write an expression for the average score per player for a team of n players who each earn S. What happens to the expression when n = 1?	nS ÷ n = S; the average equals one player's score. With n = 1 the total and average are the same.
3	The time bonus is 200 − 4s, but it can never be negative. For which whole-number values of s is the bonus positive? Explain using the expression.	200 − 4s > 0 → s < 50 (from 200 ÷ 4 = 50), so s = 0, 1, …, 49.
4	Find S for c = 15, g = 2, h = 4 with the time bonus at s = 45. Then show that changing the order of the terms does not change the answer, naming…	150 + 100 − 100 + 200 − 180 = 170. Commutative (and associative) property of addition.

Part 2: Bonus Rounds & Proofs

#	Task	Answer
5	Simplify the bonus formula 2[3(c − 2g) + 4g] − 5(c − g). What does it mean for the game?	2[3c − 6g + 4g] − 5c + 5g = 2[3c − 2g] − 5c + 5g = 6c − 4g − 5c + 5g = c + g. The bonus is just the number of coins plus gems.
6	Prove that 5(2c + 10g) − 25h and 10c + 50g − 25h are equivalent. Name each property.	5(2c + 10g) − 25h = 10c + 50g − 25h by the distributive property and the associative property of multiplication (5 · 2 = 10, 5 · 10 = 50).
7	Evaluate 200 − 4(15 − 3)² ÷ 8. A classmate got −88. Rewrite the expression with grouping symbols so that −88 would be correct.	128. The classmate computed 200 − [4(15 − 3)]² ÷ 8 = 200 − 2,304 ÷ 8 = 200 − 288 = −88.
8	The pi power-up adds π points. Prove or disprove: (a) A rational score plus π is always irrational. (b) π times a nonzero rational score is…	(a) True: if r + π = q (rational), then π = q − r would be rational, a contradiction. (b) True by the same argument with division. (c) False: π + (−π) = 0.

Part 3: Design a Balanced Game

#	Task	Answer
9	Write a general score expression with at least four terms using parameters, then choose values.	Answers vary.
10	Find two different strategies that tie. Then write the condition (an equation in your parameters) that makes trading one item for another give the…	Answers vary. Example condition: trading 1 gem for k coins keeps the score the same when q = kp.
11	Write a nested bonus-round expression and simplify it, naming each property.	Answers vary.

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Original scoring expression has at least three terms including a penalty, and it is evaluated and simplified correctly. Goes beyond the minimum with a creative, realistic choice.	Original scoring expression has at least three terms including a penalty, and it is evaluated and simplified correctly.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Pitch shows the scoring rules, the score expression with each part explained, and sample scores for two players. Polished, organized, and visually engaging.	Pitch shows the scoring rules, the score expression with each part explained, and sample scores for two players. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.